

EMPLOYEE BURNOUT & ATTRITION RISK ANALYTICS SYSTEM

1. Project Overview

The **Employee Burnout & Attrition Risk Dashboard** is a data analytics project focused on identifying employees at high risk of leaving the organization and estimating the financial impact of attrition.

Unlike traditional HR analysis that relies on basic metrics, this project uses **burnout-driven behavioural patterns and predictive modelling** to identify risk early.

The project follows a structured **3-stage pipeline**:

- **Phase 1: Data Analysis & Feature Engineering (Python - Jupyter Notebook)**

Cleaned and transformed HR dataset. Created key features such as:

- Burnout Score (0–100)
- Attrition Probability
- Revenue at Risk

This stage converts raw employee data into meaningful risk indicators.

- **Phase 2: Data Management (PostgreSQL)**

Processed dataset stored in PostgreSQL to ensure structured storage and efficient querying.

SQL used for:

- Risk-based filtering
- Department-level analysis
- Data validation

- **Phase 3: Business Intelligence (Power BI Dashboard)**

Built an interactive dashboard for **single-view risk monitoring**.

HR teams can instantly identify:

- High-risk employees
- Financial impact
- Required actions

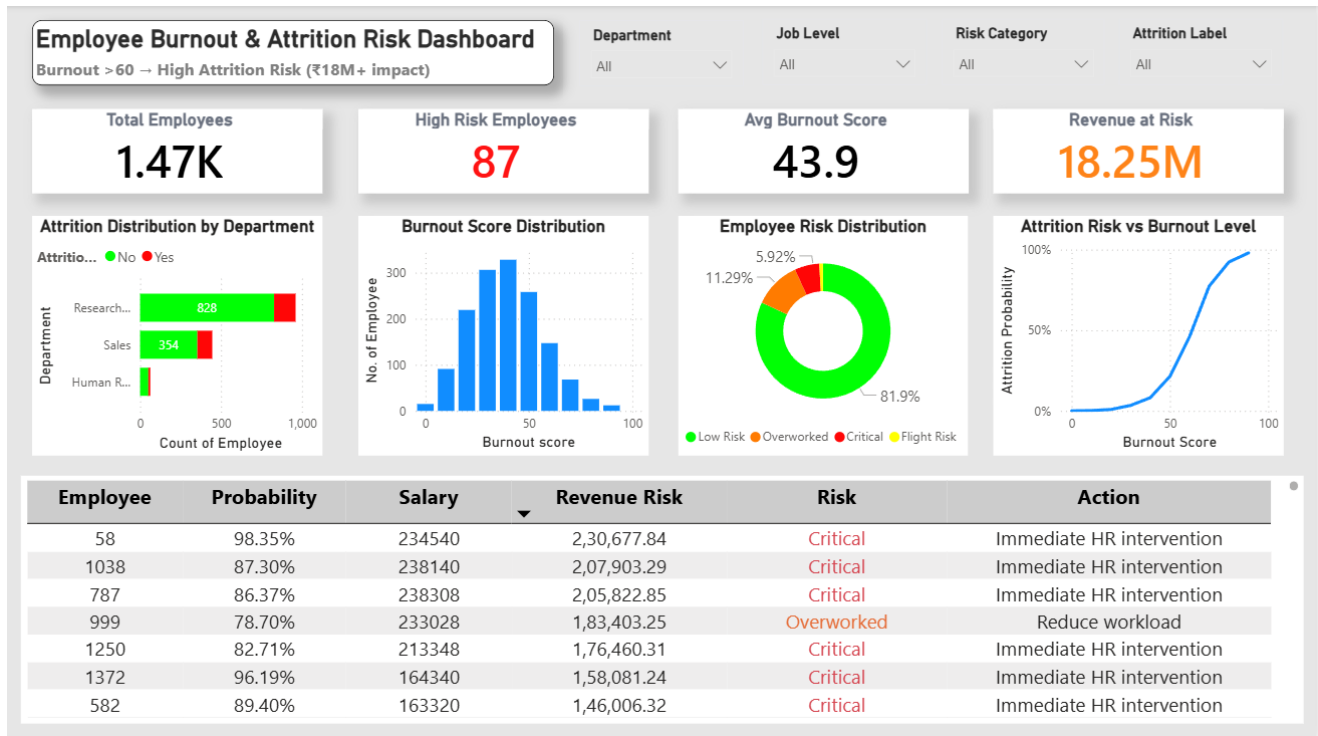


Figure 1: Final Interactive Employee Burnout Dashboard Interface

2. Technology Stack

This project utilizes a robust multi-tool architecture to ensure high-performance data processing and interactive visualization:

2.1 Jupyter Notebook (Python 3.x):

- Used for:
 - Data Cleaning
 - Feature engineering
 - Predictive modelling
- Libraries:
 - Pandas
 - NumPy
 - Scikit-learn

```
import numpy as np

df.loc[df['Attrition'] == 1, 'work_hours_per_week'] += np.random.randint(5, 15, size=sum(df['Attrition']==1))

df.loc[df['Attrition'] == 1, 'meetings_per_week'] += np.random.randint(3, 10, size=sum(df['Attrition']==1))

df.loc[df['Attrition'] == 1, 'engagement_score'] -= np.random.randint(2, 5, size=sum(df['Attrition']==1))
df['engagement_score'] = df['engagement_score'].clip(lower=1)

df.loc[df['Attrition'] == 1, 'leave_days_last_3_months'] += np.random.randint(2, 6, size=sum(df['Attrition']==1))
```

Figure 2: Integrated Technology Stack Jupyter

2.2 PostgreSQL (pgAdmin 4):

Used as relational database

- Stored processed employee dataset

Performed:

- Aggregations
- Filtering
- Validation

```
--Q1. How many high-risk employees?

SELECT risk_category, COUNT(*)
FROM "hr-employee"
GROUP BY risk_category
ORDER BY COUNT(*) DESC;

--Q2. Total revenue at risk?

SELECT SUM(revenue_at_risk) AS total_revenue_risk
FROM "hr-employee";
```

Figure 3: Integrated Technology Stack PostgreSQL

2.3 Power BI Desktop: Used for dashboard creation

Built:

- KPI cards
- Risk distribution visuals
- Department analysis

Used DAX for dynamic calculations

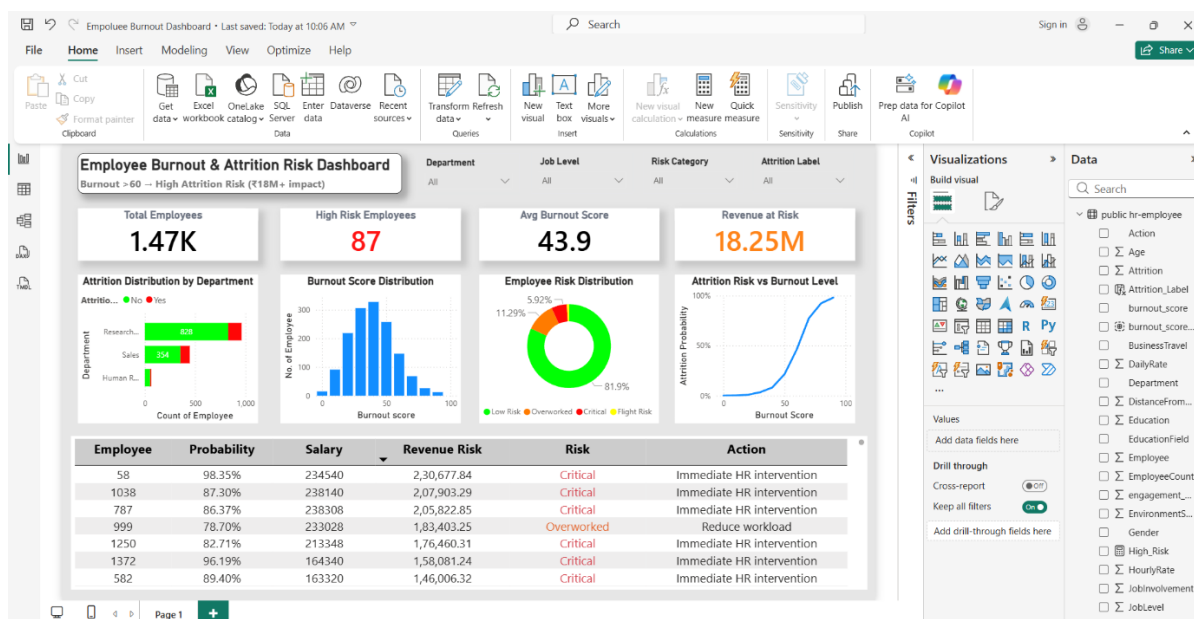


Figure 4: Integrated Technology Stack POWER BI

3. Dataset Summary

- **Dataset:** IBM HR Employee Dataset
- **Total Employees:** ~1470
- **Granularity:** Employee-level

Key Parameters:

- Burnout-related features:
 - Work hours
 - Meetings
 - Leave balance
 - Engagement score
- Target:
 - Attrition (Yes/No)

4. Methodology

The development of the employee burnout & attrition risk analytics system followed a systematic four-step analytical workflow:

4.1 Step 1: Data Acquisition & Pre-processing (Jupyter Notebook):

- Cleaned dataset
- Created:
 - Burnout Score (scaled 0–100)
 - Attrition Probability (ML model)
 - Revenue at Risk
- Categorized employees:
 - Low Risk
 - Overworked
 - Flight Risk
 - Critical

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```

Figure 5: Data Cleaning and Feature Engineering in Python

4.2 Step 2: Database Schema Design (PostgreSQL):

- Imported processed dataset
- Used SQL for:
 - High-risk employee extraction
 - Department-wise insights
 - Revenue aggregation

```
--Q3. Which department is most risky?

SELECT "Department", SUM(revenue_at_risk) AS total_risk
FROM "hr-employee"
GROUP BY "Department"
ORDER BY total_risk DESC;

--Q4. Top 10 employees to save?

SELECT "EmployeeNumber", attrition_probability, revenue_at_risk, risk_category
FROM "hr-employee"
ORDER BY revenue_at_risk DESC
LIMIT 10;
```

Figure 6: Relational Database Management in PostgreSQL

4.3 Step 3: Business Logic & Metric Development (Power BI - DAX):

- Created interactive dashboard with:
 - **KPI Cards**
 - Total Employees
 - High Risk Employees
 - Avg Burnout Score
 - Revenue at Risk
 - **Charts**
 - Attrition Distribution by Department
 - Burnout Score Distribution
 - Risk Category Distribution
 - Burnout vs Attrition Trend

- **Table**

Employee-level risk + recommended actions

Step 4: UI/UX Optimization (Theme & Layout):

- Light professional theme
- Clean spacing
- Focus on readability
- Highlighted critical metrics using colours

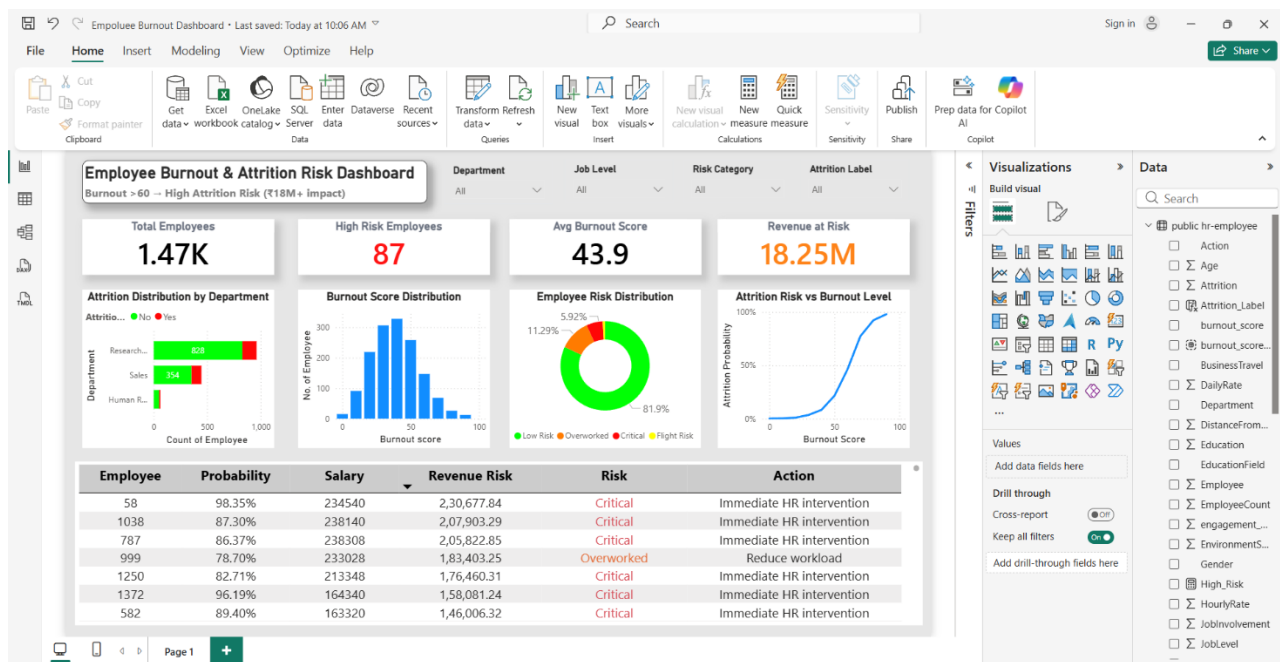


Figure 7: UI/UX Optimization using Cross-Filtering for Deep Dive Analysis

5. Dashboard Visuals & Analysis

This section illustrates the final output of the analytical pipeline. The dashboard is divided into three primary logical zones to provide a 360-degree view of employee attrition risk:

5.1 Attrition Distribution by Department

- Research and Sales departments show the highest number of employees and also higher attrition cases
- Sales has a higher proportion of attrition compared to its size, indicating pressure-driven exits
- HR department shows minimal attrition, suggesting lower workload or better balance

Attrition Distribution by Department

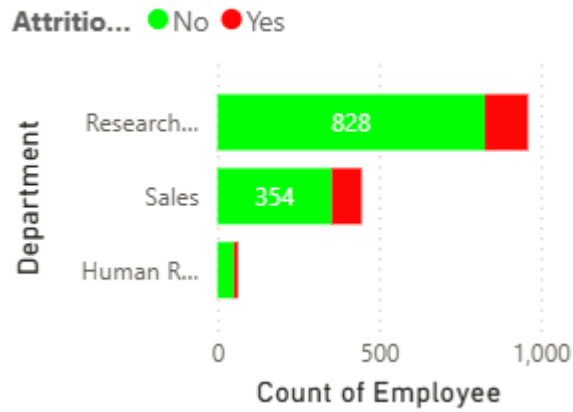


Figure 8: Attrition Distribution by Department

5.2 Burnout Score Distribution:

- Majority of employees fall in the **40–60 burnout range**
- A noticeable portion exists above **60**, where risk sharply increases
- Very few employees are in extremely low burnout range

Burnout Score Distribution

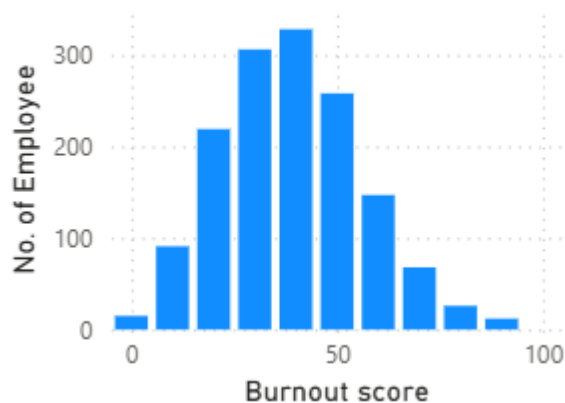


Figure 9: Burnout Score Distribution

5.3 Employee Risk Distribution:

- Around **80% employees are Low Risk**
- A smaller segment falls under:
 - Overworked
 - Critical
 - Flight Risk

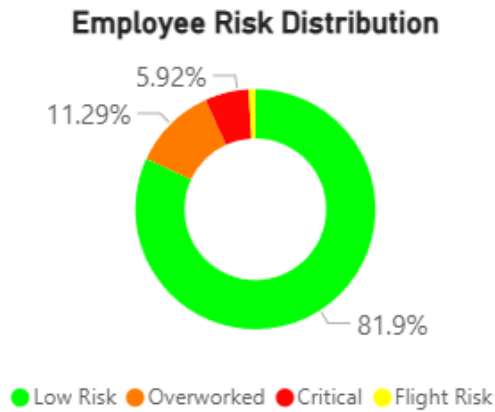


Figure 10: Employee Risk Distribution

5.4 Attrition Risk vs Burnout Level:

- Attrition probability increases **slowly till burnout ~50**
- After **60**, risk rises sharply (almost exponential trend)

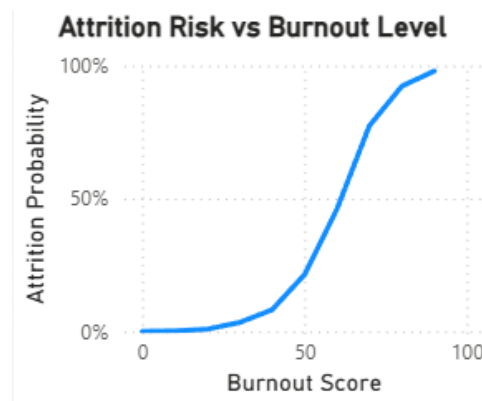


Figure 11: Attrition Risk vs Burnout Level

5.5 High-Risk Employee Analysis (Table):

- Employees with:
 - High attrition probability
 - High salary
 - High revenue risk are clearly identified
- Majority of top-risk employees fall under **Critical category**

Employee	Probability	Salary	Revenue Risk	Risk	Action
58	98.35%	234540	2,30,677.84	Critical	Immediate HR intervention
1038	87.30%	238140	2,07,903.29	Critical	Immediate HR intervention
787	86.37%	238308	2,05,822.85	Critical	Immediate HR intervention
999	78.70%	233028	1,83,403.25	Overworked	Reduce workload
1250	82.71%	213348	1,76,460.31	Critical	Immediate HR intervention
1372	96.19%	164340	1,58,081.24	Critical	Immediate HR intervention
582	89.40%	163320	1,46,006.32	Critical	Immediate HR intervention

Figure 12: High-Risk Employee Analysis (Table)

6.Business Recommendations

Based on the data-driven insights generated by the dashboard, the following attrition risk mitigation strategies are recommended:

6.1 Early Risk Intervention (Burnout Threshold)

- Flag employees with burnout score ≥ 60 as priority risk
- Set automated alerts for HR

6.2 Target High-Risk Employees First

- Prioritize employees with:
 - High attrition probability
 - High revenue at risk

Action

- 1:1 HR intervention (Critical)
- Immediate workload correction (Overworked)
- Retention offer / role change (Flight Risk)

Impact: Maximum risk reduction with minimum effort

6.3 Department-Level Workload Optimization

- Focus on Sales and Research teams

Action:

- Rebalance workload
- Reduce overtime
- Improve team distribution

Impact: Lower burnout at source (not just symptoms)

6.4 Continuous Monitoring System

- Integrate dashboard into monthly HR review

Track:

- Burnout trends
- Risk category movement
- Revenue at risk

Impact: Shift from reactive → proactive HR

6.5 Retention Strategy for Key Employees

- Identify high-salary + high-risk employees

Action:

- Incentives
- Career growth plans
- Flexible work policies

Impact: Protect high-value talent

6.6 Policy-Level Changes

- Limit excessive meetings
- Enforce work-hour boundaries
- Improve engagement programs

Impact: Long-term burnout reduction

7. Conclusion

This project demonstrates how data analytics can be used to move HR decision-making from intuition-based to data-driven.

By combining **Python (feature engineering)**, **SQL (data management)**, and **Power BI (visualization)**, the system successfully:

- Identifies employees at high risk of attrition
- Quantifies potential financial impact (₹18M+)
- Provides actionable recommendations for HR

The key finding is that **burnout is a strong predictor of attrition**, with risk increasing sharply beyond a threshold level.

This solution enables organizations to:

- Detect risk early
- Take targeted actions
- Reduce attrition-related losses

Overall, This system enables proactive intervention, reducing attrition before it impacts business performance.